DEPARTMENT OF MATHEMATICS AND STATISTICS
UNIVERSITY OF MASSACHUSETTS, AMHERST

# ALGEBRA ADVANCED EXAM. AUGUST 2026

**Passing Standard:** The passing standard is 70% with essentially correct solutions to five problems.

**Instructions:** Show all your work and justify your answers carefully. All rings are commutative and contain the identity, and all ring homomorphisms preserve the identity.

## PART I. GROUP THEORY

**1.** Let $p$ be a prime number and let $G$ be a non-abelian group of order $p^3$. Denote by $Z(G)$ the center of $G$ and by $[G,G]$ the commutator subgroup of $G$, i.e. the subgroup generated by all $xyx^{-1}y^{-1}$ with $x, y \in G$.

(a) Show that $\#Z(G) = p$ and that $Z(G) = [G,G]$.

(b) Determine, with proof, the number of conjugacy classes of $G$.

*Please work out each problem on a separate sheet of paper.*

**2.** Let $G$ be a finite group.

(a) Let $H \subsetneq G$ be a proper subgroup. Show that $\bigcup_{g\in G} gHg^{-1} \neq G$.

(b) Suppose $G$ acts transitively on a set $X$ with $\#X \geq 2$. Show that there exists $g \in G$ such that $g(x) \neq x$ for every $x \in X$.

(c) Show that the conclusion of part (a) may fail when $G$ is infinite.
*[Hint: take $G = \mathrm{GL}_2(\mathbb{C})$ and let $H$ be the subgroup of upper triangular matrices.]*

## PART II. COMMUTATIVE ALGEBRA

*Note: every ring has an identity, and every ring homomorphism preserves the identity.*

**3.** Let $A$ be a ring and let $M$ be a finitely generated $A$-module.

(a) Let $\varphi\colon M \to M$ be a surjective homomorphism of $A$-modules. Show that $\varphi$ is injective.
*[Hint: make $M$ into an $A[t]$-module by letting $t$ act as $\varphi$, and imitate the determinant trick used in the proof of Nakayama's lemma.]*

(b) Give an example of a ring $A$, a finitely generated $A$-module $M$, and an *injective* homomorphism of $A$-modules $M \to M$ that is *not* surjective.

(c) Let $N$ be an $A$-module such that $M \simeq M \oplus N$ as $A$-modules. Show that $N = 0$.

**4.** Let $R = \mathbb{Z}[\sqrt{-3}] \subset \mathbb{C}$, let $K = \mathbb{Q}(\sqrt{-3})$ be its field of fractions, and let

$$I = \left(2,\ 1 + \sqrt{-3}\right) \subseteq R.$$

Recall that a non-zero ideal $J$ of $R$ is called *invertible* if there is a finitely generated $R$-submodule $J' \subseteq K$ with $JJ' = R$.

(a) Show that $R$ is not integrally closed in $K$.

(b) Show that $I$ is a maximal ideal of $R$ and that $I^2 = 2I$.

(c) Deduce that $I$ is not invertible, and that the ideal $(2)$ of $R$ is *not* a product of prime ideals of $R$.

*Please work out each problem on a separate sheet of paper.*

**5.** Let $A \neq 0$ be a Noetherian ring and let $N = \sqrt{0}$ be its nilradical.

(a) Show that every proper radical ideal of $A$ is an intersection of finitely many prime ideals. Deduce that $A$ has only finitely many minimal primes.

(b) Show that $N$ is nilpotent, i.e. that $N^m = 0$ for some integer $m \geq 1$.

(c) Exhibit a ring with infinitely many minimal primes, and prove that your example has this property.

## PART III. FIELD THEORY AND GALOIS THEORY

**6.** Let $p$ be a prime and $n \geq 1$ an integer. Denote by $N_p(n)$ the number of monic irreducible polynomials of degree $n$ in $\mathbb{F}_p[x]$, and by $\mu$ the Möbius function.

(a) Show that $x^{p^n} - x$ is the product of all monic irreducible polynomials of $\mathbb{F}_p[x]$ whose degree divides $n$, each occurring exactly once.

(b) Deduce that $\sum_{d|n} d\, N_p(d) = p^n$, and hence that

$$N_p(n) = \frac{1}{n} \sum_{d|n} \mu(d)\, p^{\,n/d}.$$

Compute $N_2(6)$.

(c) Using the formula in (b), show that $N_p(n) > 0$ for all $p$ and all $n \geq 1$.

**7.** Let $p$ be an odd prime, let $\zeta = e^{2\pi i/p}$, and set $p^* = (-1)^{(p-1)/2}p$. For $a \in \mathbb{Z}$ with $p \nmid a$ let $\left(\frac{a}{p}\right) \in \{\pm 1\}$ denote the Legendre symbol, which is $+1$ if $a$ is a square modulo $p$ and $-1$ otherwise.

(a) Show that $\mathbb{Q}(\zeta)$ contains exactly one subfield $F$ with $[F : \mathbb{Q}] = 2$.

(b) Let $g = \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) \zeta^a$. Show that $g^2 = p^*$, and conclude that $F = \mathbb{Q}(\sqrt{p^*})$.

(c) Deduce that every quadratic extension of $\mathbb{Q}$ is contained in a cyclotomic field.

*Please work out each problem on a separate sheet of paper.*

**8.** Let $L/K$ be a finite extension of fields.

(a) Suppose $L = K(\alpha)$ for some $\alpha \in L$. Show that there are only finitely many fields $F$ with $K \subseteq F \subseteq L$.

(b) Conversely, suppose that there are only finitely many fields $F$ with $K \subseteq F \subseteq L$. Show that $L = K(\alpha)$ for some $\alpha \in L$.

(c) Let $p$ be a prime, let $s, t$ be algebraically independent indeterminates over $\mathbb{F}_p$, and set

$$K = \mathbb{F}_p(s^p, t^p) \subset L = \mathbb{F}_p(s, t).$$

Show that $[L : K] = p^2$ and that $L/K$ has infinitely many intermediate fields.

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